

MARKET CASE / 2026

Market size, capture and unit economics

A source-linked commercial case for an Indian adaptive acoustic payload. Global references are converted to INR; Indian demand is built from named programmes and estimated deployable units.

INR 452.3 cr	INR 31.9 cr	INR 6.61 cr	39.7%
2025 narrow global TAM	5-year India SAM	5-year staged SOM	Target gross margin

TAM INR 452.30 crore
Global side-scan sonar

SAM INR 31.90 crore
5-year India serviceable demand

SOM INR 6.61 crore
5-year staged revenue

Decision summary

Decision metric	Position
Initial customer	Universities, AUV/ROV teams, inland-water and port inspection pilots
Target offer	INR 1.95 lakh ex-GST evaluation unit with local integration
Scale constraint	Qualification, calibrated acoustic evidence, sea trials and procurement cycles
Commercial wedge	On-device adaptive transmit intelligence with transparent physics and energy guards

TAM uses the 2025 USD 47.25 million side-scan sonar estimate and INR 95.7245/USD. The category is intentionally narrower than broad marine electronics or ocean-technology markets.

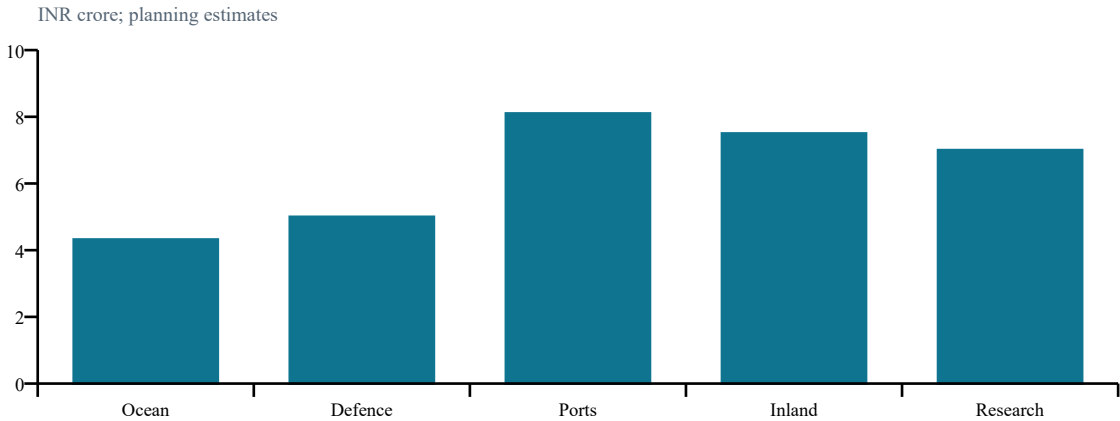
02 / INDIA SERVICEABLE MARKET

Bottom-up SAM from deployable units

Programme budgets establish active demand environments; they are not treated as revenue. The SAM estimates the number of payload opportunities that SeaNergy can technically and commercially serve over five years.

Customer group / programme context	Units	ASP	5-year value
MoES / NIOT and Deep Ocean Mission research payloads	24	INR 18 lakh	INR 4.32 cr
Navy, Coast Guard, iDEX and ADITI pilots	20	INR 25 lakh	INR 5.00 cr
Major ports and coastal infrastructure inspection	45	INR 18 lakh	INR 8.10 cr
Inland waterways, dams and reservoirs	50	INR 15 lakh	INR 7.50 cr
Universities, survey labs and vehicle developers	35	INR 20 lakh	INR 7.00 cr
Total	174	-	INR 31.92 cr

Five-year India SAM by customer group



Programme evidence

Programme	Published scale	Commercial relevance
Deep Ocean Mission	INR 4,077 crore approved outlay	Ocean technology, vehicles, sensors and deep-ocean capability
ADITI under iDEX	INR 750 crore; grants up to INR 25 crore	Indigenous critical defence technology and prototype adoption
Sagarmala	839 projects; about INR 5.8 lakh crore investment	Ports, coastal assets, waterways and inspection settings

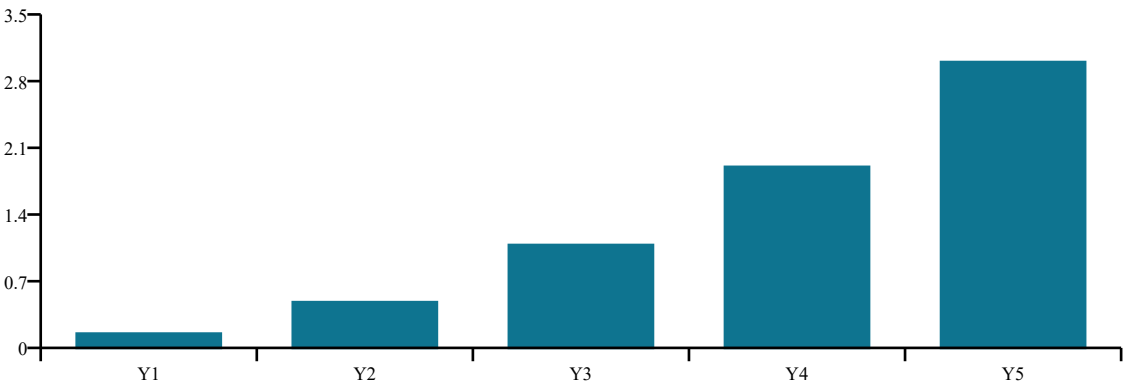
03 / CAPTURE AND ECONOMICS

Qualification-led five-year capture

35 units	INR 6.61 cr	20.7%	INR 77,352
Cumulative SOM	Cumulative revenue	SOM / SAM	Gross profit per unit

Staged revenue capture

INR crore; qualification and field evidence gate the ramp



Unit economics

COGS element	INR / unit
Electronics and PCB	26,148
Enclosure and sealed integration	18,500
Transducers	36,000
Battery and power	8,500
Assembly and calibration	17,000
QA and warranty reserve	8,000
Packaging and field kit	3,500
Total COGS	1,17,648
Target selling price, ex-GST	1,95,000
Gross profit	77,352
Gross margin	39.7%

The model treats engineering labor for repeatable assembly and calibration as COGS. Product development, calibrated lab equipment and field trials remain programme expenses.

04 / COMPETITION AND ENTRY

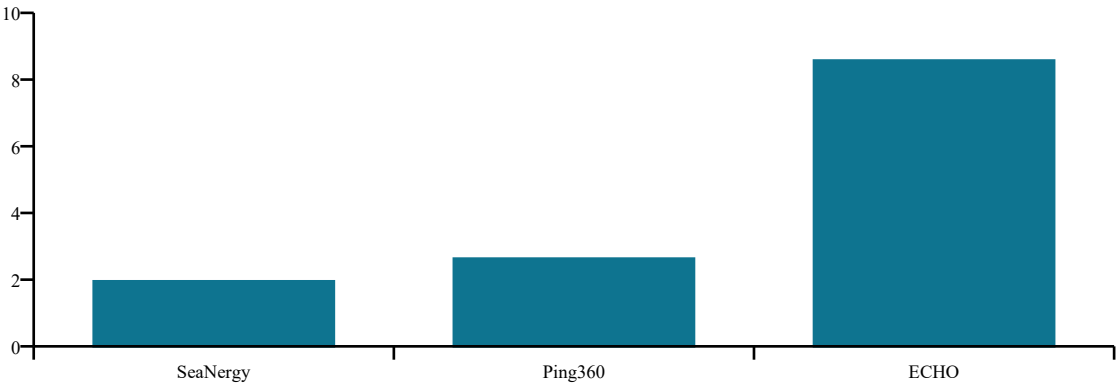
Win the adaptive-controller wedge first

SeaNergy enters below imported imaging-sonar price points and avoids claiming full survey-system parity. The first sale is an evaluation payload with transparent waveform decisions, local integration and an upgrade path to calibrated receive and 100 W drive.

Reference	Public price	INR conversion	What it proves
SeaNergy target unit	INR 1.95 lakh	INR 1.95 lakh	Local target price with adaptive controller
Blue Robotics Ping360	USD 2,750	INR 2.63 lakh	Integrated compact scanning sonar benchmark
Sonoptix ECHO	From USD 8,950	INR 8.57 lakh	Entry multibeam imaging-sonar benchmark

Public-price positioning

INR lakh before taxes and import costs; procurement benchmark, not feature equivalence



Route to market



Evidence required at each step

Evaluation	Paid pilot	Field payload	Platform / OEM
Decision trace, energy and application	Tank calibration and mission metric	Calibrated source level, ingress and integration	Stable interfaces, manufacturing controls and service

Sources and audit trail

Programme references: PIB Deep Ocean Mission and ADITI releases; Sagarmala project dashboard. Market references: DIResearch and GlobalInfoResearch. Product prices: Blue Robotics. FX: RBI reference-rate archive via MSEI, 11 September 2026. Full URLs and assumptions are recorded in market/sources.md and cost-model.xlsx.